Operationalisation of Key Constructs

This study examines how reproducible research practices can be designed, evaluated, and sustained within an applied university research environment. To empirically investigate this, theoretical constructs derived from the selected theories are translated into observable and measurable elements. This operationalisation ensures that abstract concepts are examined through concrete evidence rather than assumed compliance.

Each construct is operationalised using a consistent structure consisting of a construct and dimension, an operational criterion, observable indicators, and corresponding measurement instruments. Data collection uses interviews, surveys, and document analysis in parallel to allow comparison between reported practice, perceived quality, and documented evidence.

The selected theories address complementary aspects of reproducible research, covering framework development, reuse of outputs, traceability of decisions, system level alignment, and long term embedding of practices.

Design Science Research

Construct and dimension
Iterative development and evaluation of a reproducible research framework.

Operational criterion
The research framework is developed, tested, and adjusted through repeated use in real student projects rather than designed or evaluated only once.

Indicators
The framework is used during multiple phases of a project.
Moments of evaluation or reflection on the framework are documented.
Feedback or observed problems lead to visible changes in how the framework is applied.

Operationalisation into measurement instruments
This construct is examined by reconstructing how the framework was used over the course of projects. Interviews with students and supervisors focus on when the framework influenced decisions and whether it was adapted during the project. Surveys measure how often the framework was used at different project stages. Document analysis checks whether changes, evaluations, or adjustments to the framework are visible in project materials. Together, these methods assess whether the framework followed an iterative build and test logic rather than a one time implementation.

FAIR Principles

Construct and dimension
Reusability and accessibility of research outputs.

Operational criterion
Research outputs are structured so that a new team can locate, understand, and reuse them without relying on informal explanations from the original researchers.

Indicators
Consistent file and folder structures across projects.
Clear documentation of project context, assumptions, and scope.
Stable and accessible storage locations for research materials.
Guidance explaining how outputs can be continued or reused.

Operationalisation into measurement instruments
This construct is assessed by comparing perceived reusability with observable evidence. Interviews and surveys ask participants whether they believe their outputs can be reused and why. Document analysis verifies whether required elements such as structure, context, and accessibility are actually present. Comparing perceptions with documented reality allows identification of gaps that limit reuse across cohorts.

Provenance Theory

Construct and dimension
Traceability of research decisions, data, and conclusions.

Operational criterion
Research outputs contain enough information to allow an independent person to reconstruct how conclusions were derived from data and methods.

Indicators
Clear links between data sources, analysis steps, and conclusions.
Documented reasoning behind key methodological or analytical choices.
Records of changes to scope, methods, or assumptions during the project.

Operationalisation into measurement instruments
This construct is examined by testing whether reconstruction is possible in practice. Interviews explore whether participants can explain how conclusions were reached and where undocumented decisions occurred. Surveys capture perceptions of completeness in decision documentation. Document analysis traces whether links between data, analysis, and

conclusions are explicitly visible. This approach verifies traceability based on evidence rather than intention.

Work System Theory

Construct and dimension
Alignment of roles, processes, tools, and information within the research system.

Operational criterion
The organisation of student research supports consistent documentation and reproducible practices across projects and cohorts.

Indicators
Shared understanding of documentation expectations among students and supervisors.
Clear allocation of responsibility for documentation quality.
Consistent use of tools, templates, and repositories.
Defined handover moments between cohorts.

Operationalisation into measurement instruments
This construct is examined at the system level. Interviews and focus groups identify mismatches between expected and actual documentation practices across roles. Surveys assess how clearly responsibilities and expectations are understood. Document analysis verifies whether shared tools, templates, and handover materials exist and are used consistently. This allows identification of systemic issues that undermine reproducibility beyond individual behaviour.

Normalization Process Theory

Construct and dimension
Embedding of reproducible research practices into routine work.

Operational criterion
Reproducible research practices are understood, accepted, and routinely applied as part of normal research activities.

Indicators
The framework is used during active decision making, not only during final reporting.
Supervisors actively reinforce documentation and reproducibility during guidance and assessment.
Use of the framework continues across multiple cohorts.

Operationalisation into measurement instruments
This construct is examined by assessing whether practices are part of everyday research work. Interviews and focus groups explore how participants experience the framework and whether it feels integral or additional. Surveys measure how regularly the framework is used during ongoing work. Document analysis checks whether framework related artefacts appear throughout project timelines and across cohorts. This shows whether practices are becoming routine rather than isolated interventions.

Summary of Operationalisation Approach

By translating each theoretical construct into clear criteria, observable indicators, and concrete measurement instruments, this study establishes a direct link between theory, data collection, and analysis. The use of interviews, surveys, and document analysis in parallel enables triangulation and reduces reliance on self reported claims. This operationalisation provides a robust empirical foundation for answering the research questions and for defining justified requirements and criteria in the solution development phase.